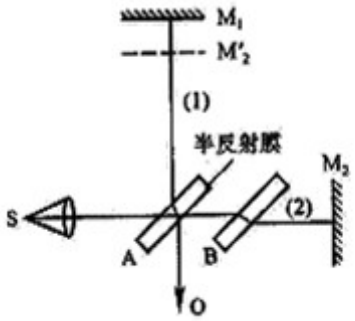
* 6

1

1 A B A M_1 M_2 M_2

M_1 10^{-4} mm 10^{-5}



mm M_1 M_2

S A (1) (2) M_1 M_2

A O B (1) (2)

M_1 M_2 A

O A

M_1 A M_2 2’ M_1 M_2 M_1 M_2' M_1

M_2' “ ” M_2' (M_1 M_2') M_1 M_2'

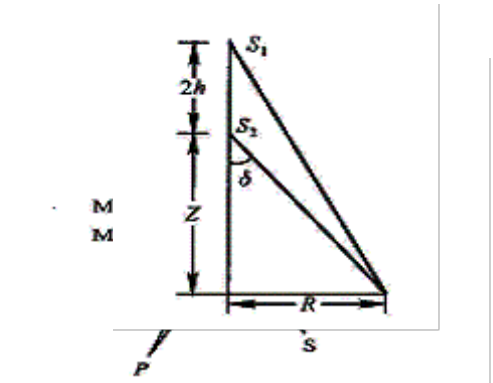
2

S M_1 M_2'

S_1 S_2 (2) (M_1 M_2') h

S_1 S_2 2h M_1 M_2' (M_2)

P S_1 S_2 P



P S_1 S_2 (P) Δ P S_1 S_2

S_1 S_2 3 Δ

$$\Delta=\sqrt{(Z+2h)^2+R^2}-\sqrt{Z^2+R^2}=\sqrt{Z^2+R^2}\left[\left(1+\frac{4Zh+4h^2}{Z^2+R^2}\right)^{\frac{1}{2}}-1\right]$$

$$\Delta = \sqrt{Z^2+R^2}\left[\frac{1}{2}\left(\frac{4Zh+4h^2}{Z^2+R^2}\right)-\frac{1}{8}\left(\frac{4Zh+4h^2}{Z^2+R^2}\right)^2+\cdots\right]$$
$$\approx \frac{2hZ}{\sqrt{Z^2+R^2}}\left[\frac{Z^3+ZR^2+R^2h-2h^2Z-h^3}{Z(Z^2+R^2)}\right]$$
$$= 2h\cos\delta\left[1+\frac{h}{Z}\sin^2\delta-\frac{2h^2}{Z^2}\cos^2\delta-\frac{h^3}{Z^3}\cos^2\delta\right]$$

h<<Z

$$\Delta = 2h\cos\delta\left(1+\frac{h}{Z}\sin^2\delta\right) \quad (1)$$

(1) $\delta_{=0}$ $\Delta_1 = 2h_1 = m\lambda$

$$\Delta_2 = 2h_2 = (m+n)\lambda$$

$$\Delta h = h_2 - h_1 = \frac{1}{2}(\Delta_2 - \Delta_1) = \frac{1}{2}n\lambda \quad (2)$$

M_1 M_2 “ ” “ ” “ ”

“ ” ( ) (2) $\Delta h = \frac{1}{2}n\lambda$ M_1

Δh ( )

M_1 M_2'

M_1 M_2' “ ”

3

λ_1 λ_2 λ_1 λ_2

$L = m\lambda_1 = (m+\frac{1}{2})\lambda_2$ (m) λ_1

λ_2

M_1 ( ) λ_1 λ_2

M_1 $L+\Delta L = (m+\Delta m)\lambda_1 = (m+\Delta m+\frac{1}{2})\lambda_2$ ( )

$$\Delta L = \Delta m\cdot\lambda_1 = (\Delta m+1)\lambda_2$$

$$\Delta\lambda = \frac{\lambda_1\lambda_2}{\Delta L} = \frac{\lambda^2}{\Delta L} \qquad (3)$$

$$\Delta\lambda = |\lambda_1 - \lambda_2|$$

$$\lambda = \frac{\lambda_1+\lambda_2}{2}$$

(3)

4

i=0 d

L=2d d d’ $2d'$ L_m c

t_m

“ ” λ_0

$\Delta\lambda$ “ ” $\lambda_0 - \frac{\Delta\lambda}{2}$ $\lambda_0 + \frac{\Delta\lambda}{2}$ d

$\lambda_0 + \frac{\Delta\lambda}{2}$ $\lambda_0 - \frac{\Delta\lambda}{2}$

d

$2d' = L_m$ L_m λ_0 $\Delta\lambda$

$$L_m = \frac{\lambda_0^2}{\Delta\lambda} \qquad (4)$$

$\Delta\lambda$ t_m

$$t_m = \frac{L_m}{c} = \frac{\lambda_0^2}{c\Delta\lambda} \qquad (5)$$

589.0nm 589.6nm 2cm 632.8nm $\Delta\lambda$

10^{-14}~10^{-7}nm $\Delta\lambda$ λ

5 ( )

(1).

d=0 M_1 M_2'

$\lambda/2$($\lambda/2$) d

d=0

(2)

M_1 A n l

$\Delta L = 2l(n-1)$

M_1 A d $d = \frac{\Delta L}{2}$ d *l*

d=l(n-1) (6)

n

He-Ne Na ( )

1.

(1) M_1 M_2 (M_1// M_2)

He-Ne M_2 M_2 (M_1)

M_1 M_2

(2) M_2

(3) “ ” “ ”

2 He-Ne

: M_1 h “ ” “ ”50 h N

500 (2)

$\Delta h = h_2 - h_1 = \frac{1}{2}(\Delta_2 - \Delta_1) = \frac{1}{2}n\lambda$

λ

3

4

5 .

M_1 ( )

1

1 M1 “ ” “ ”30 M1

	1	2	3	4	5	6

D(mm)	30.00200	30.01082	30.01970	30.02855	30.03736	30.04620

2

	1	2	3
ΔD(mm)	0.02655	0.02654	0.02650

3 λ=589.56nm

4 M1 M1

	1	2	3	4
D(mm)	30.22255	30.50900	30.79900	31.08900

5

	1	2
ΔD(mm)	0.57645	0.58000

6 d=0.60nm

2

1 M1 d=0

① M1 31.18250mm

② M1 31.17750mm

2 M1 d=0

① M1 31.18250mm

② M1 31.17710mm

3 0.01mm

N	1.50	1.54

3 He-Ne

1 M1 “ ” “ ”30 M1

	1	2	3	4	5	6
D(mm)	30.50000	30.50958	30.51905	30.52858	30.53805	30.54755

2

	1	2	3
ΔD(mm)	0.02858	0.02847	0.02850

3 He-Ne λ=633.70nm

1

① 589.56nm 589.30nm 0.04% d=0.60nm

② 1.50 1.51 0.7% 1.54

③ He-Ne 633.70nm 632.80nm 0.1%

④

2

① 30

②

3

①

②